# WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

### Guided 1 — original CHECK

Intersect term postings

Follow the second stage and explain its importance.

Return D

Inverted index: encryption→C,D and hash→D. Intersecting returns D. A search engine may rank by further signals after matching.

IMPROVE: Explain one correction to your own reasoning.





DO NOW — RECALL

# Lesson 154 · Search Engines: Crawling, Indexing and Re

Name: _____ Class: _____ Date: _____

**1. Why does a server recheck a browser form?**

**2. When can binary search be used?**

**3. What data structure follows LIFO?**

**4. What is a database foreign key?**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

**1. Model answer**

Client checks are bypassable.

**2. Model answer**

On sorted data.

**3. Model answer**

Stack.

**4. Model answer**

A reference to a key in another table.

**By the end, you can…**

- Crawler fetches pages found through links and sitemaps.
- Map each searchable term to candidate pages.
- Match query terms and rank relevant results.

**Vocabulary — know these terms**

crawler — Search agent fetching linked pages.

inverted index — Terms mapped to document identifiers.

tokenisation — Breaking text into searchable units.

ranking — Ordering candidate search results.



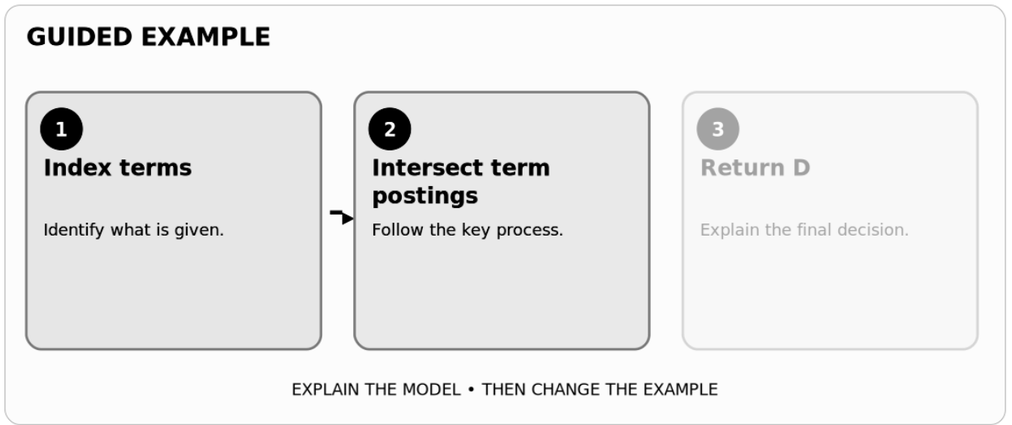


# We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Page C has 'encryption key' and Page D 'encryption hash'. Which page answers 'encryption hash'?**

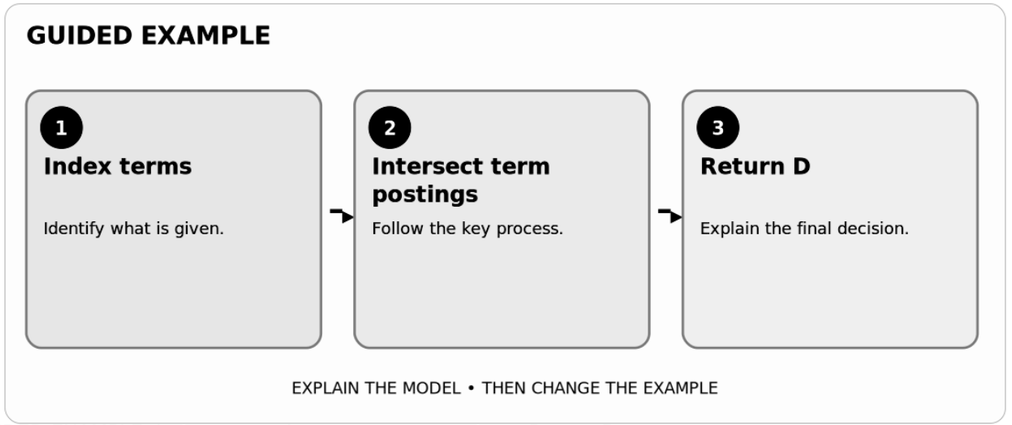
Identify the input or diagram and predict the first result.




GUIDED EXAMPLE: Index terms, Intersect term postings, Return D

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EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**Search indexing and result ranking are the same operation.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.



RHS COMPUTING
OCR H446 · LESSON 154 · DRAFT

NEW LEARNING

# Read, annotate and explain

**Discover**

Crawler fetches pages found through links and sitemaps.

**Index**

Map each searchable term to candidate pages.

**Retrieve**

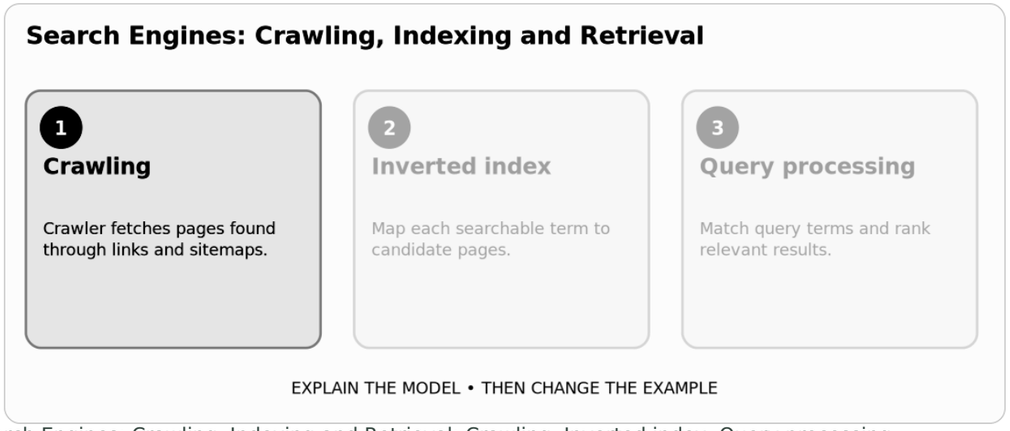
Match query terms and rank relevant results.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

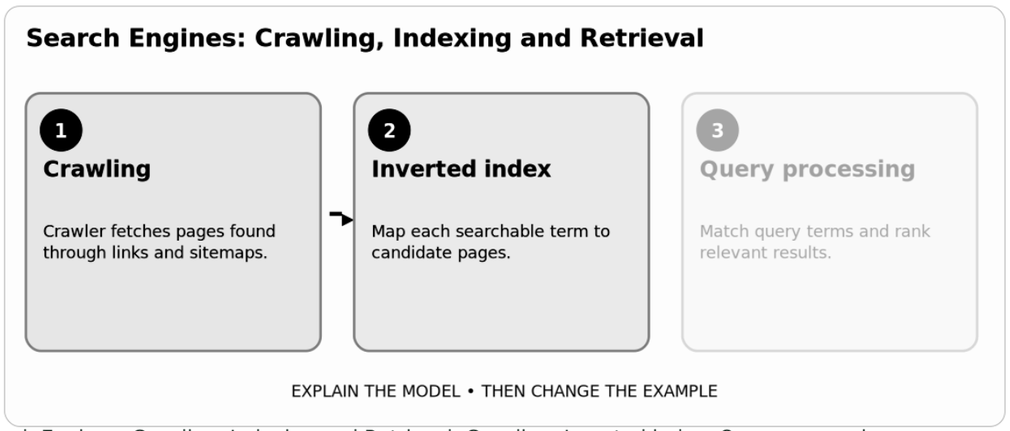
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### New Learning




Search Engines: Crawling, Indexing and Retrieval: Crawling, Inverted index, Query processing

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YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Discuss why a new page might not appear in search and why robots.txt is not a security control. [8]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Explain crawling and indexing. [6]

### Question 2

Index P='cat dog', Q='dog bird' and identify result for 'dog bird'. [6]

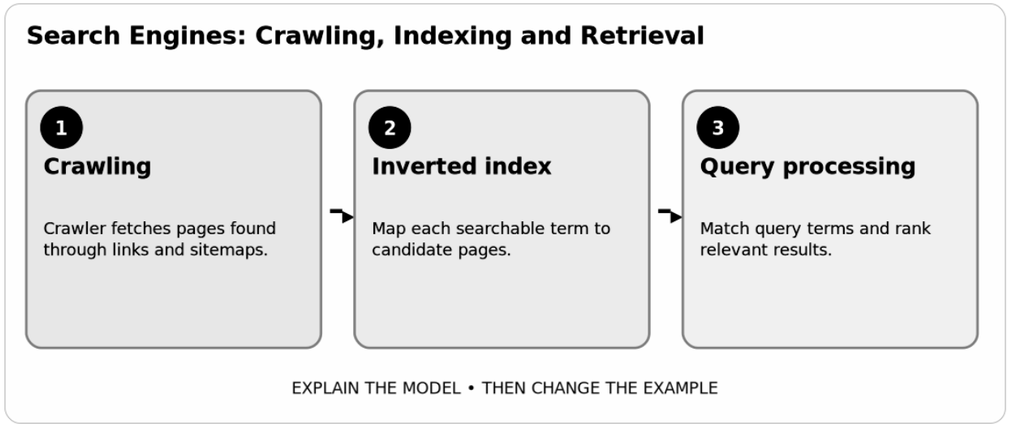




# New Learning · original lesson diagrams

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### New Learning




Search Engines: Crawling, Indexing and Retrieval: Crawling, Inverted index, Query processing

I DO — WORKED EXAMPLE

# Follow the worked method

**Page A has 'binary search' and Page B 'binary tree'. Build a small inverted index.**

**Tokenise both pages**

Identify the input or diagram and predict the first result.

**Map terms to page IDs**

Follow the second stage and explain its importance.

**Match search → A**

binary→A,B; search→A; tree→B. The query 'binary search' intersects both postings and retrieves A.





# WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

**Guided 1: Index terms Identify the input or diagram and predict the first result.**

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

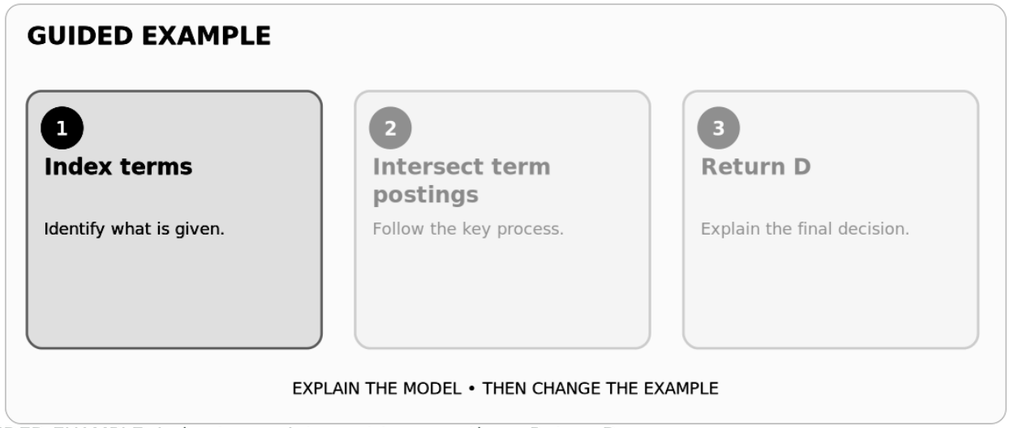
IMPROVE — Write one precise correction or stronger explanation in your own answer.

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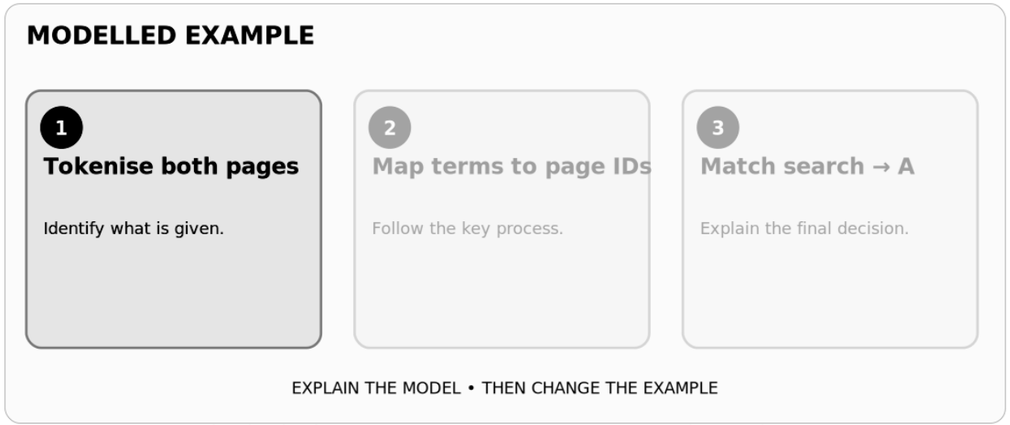


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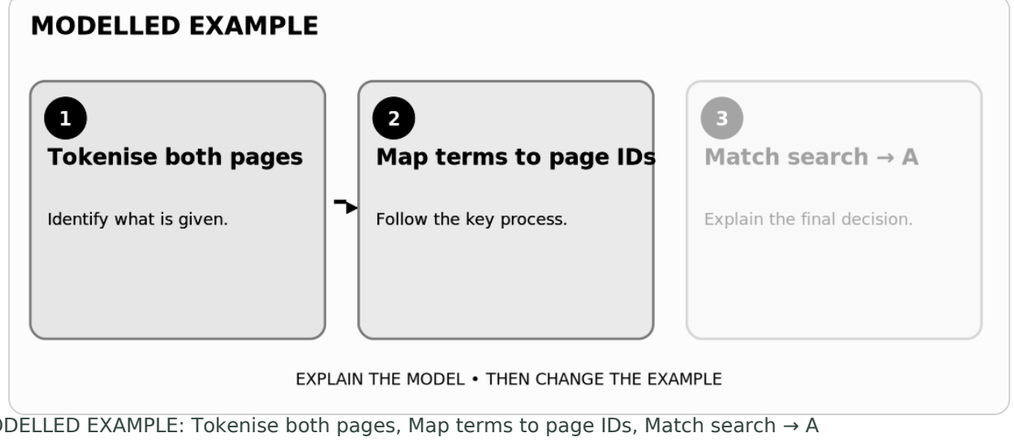
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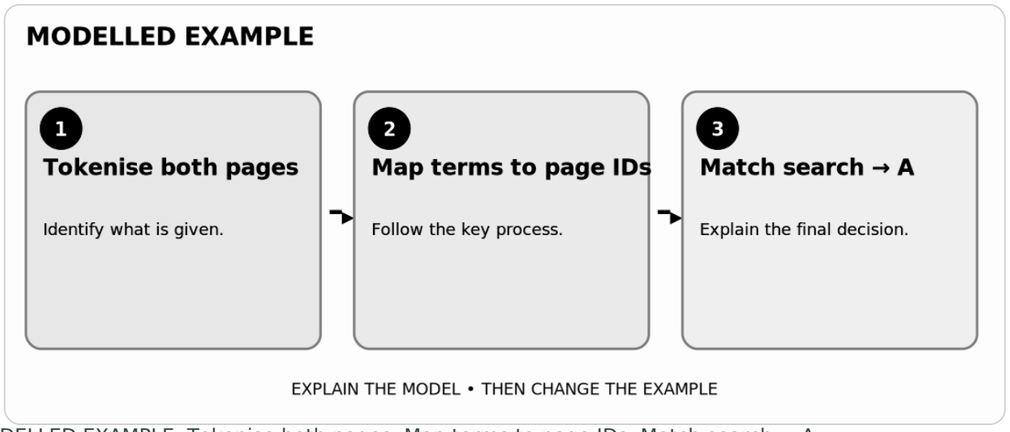

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**WE DO — GUIDED PRACTICE**

# Try → Discuss → Improve